BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The digestive system, part I: through the esophagus

The written notes for the first digestive lecture: the tube and its wall, the enteric plexuses, the peritoneum, the mouth, salivary glands, tongue, pharynx, esophagus, and the sphincters. Module 4 competencies 109 to 114, 120, and 121, with nine worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Before the competencies: the two groups of organs and the six jobs

In your course notes: Module 4 packet, Chapter 3 introduction (page 25).

Dr. Rennie's video for this part: Digestive system concept video, chapter Digestive organ overview, and GI tract core processes.

The digestive system is two groups of organs. The gastrointestinal (GI) tract, also called the alimentary canal, is one continuous tube from the mouth to the anus that runs through the thoracic and abdominal cavities: mouth, most of the pharynx, esophagus, stomach, small intestine, and large intestine. The accessory digestive organs sit beside the tube: the teeth and tongue, which touch food directly, and the salivary glands, liver, gallbladder, and pancreas, which make or store secretions and send them into the tube through ducts.

- In a living person the tract is about 5 to 7 m (16.5 to 23 feet) long, because its muscle is in a steady state of partial contraction (tonus). After death the muscle relaxes and the tube measures about 7 to 9 m (23 to 29.5 feet).

The six basic processes of the GI tract

1. Ingestion: taking food and liquid into the mouth.
2. Secretion: water, acid, buffers, and enzymes released into the lumen.
3. Mixing and propulsion: the smooth muscle of the wall contracts and relaxes, mixing food with secretions and pushing it forward.
4. Digestion: breaking food into smaller molecules. Mechanical digestion is the cutting and grinding of the teeth and the churning of the stomach and small intestine. Chemical digestion is enzymes splitting carbohydrates, lipids, proteins, and nucleic acids by hydrolysis. Vitamins, ions, water, and cholesterol can be absorbed without chemical digestion.
5. Absorption: digested molecules cross the epithelium from the lumen into the blood or lymph.
6. Defecation: eliminating what cannot be used (feces: wastes, indigestible material, bacteria, and cells shed from the lining).

Competency 109. Four-layer gut wall

In your course notes: Module 4 packet, Chapter 3, competency 109 (page 25).

Dr. Rennie's video for this part: Digestive system concept video, chapter GI wall layers and mucosa, and Epithelial types and functions.

You should be able to: Name the four layers of the alimentary canal wall from lumen outward and state the dominant tissue of each.

From the esophagus to the anal canal the wall of the alimentary canal has the same four layers. From the lumen outward: mucosa, submucosa, muscularis, serosa (or adventitia). Every region changes the details to suit its job, but the order never changes.

The mucosa, the inner lining, has three parts

1. Epithelium, in direct contact with the food. Nonkeratinized stratified squamous epithelium in the mouth, pharynx, esophagus, and anal canal, where the job is protection from rubbing. Simple columnar epithelium in the stomach and intestines, where the job is secretion and absorption. Tight junctions seal the cells together so nothing leaks between them, and the cells turn over every 5 to 7 days. Exocrine cells secrete mucus and fluid into the lumen; enteroendocrine cells secrete hormones into the blood.
2. Lamina propria: areolar connective tissue carrying the blood and lymphatic vessels that pick up absorbed nutrients. It supports the epithelium and holds mucosa-associated lymphatic tissue (MALT), the lymphatic nodules that guard the tract. MALT is heaviest in the tonsils, the small intestine, the appendix, and the large intestine.
3. Muscularis mucosae: a thin layer of smooth muscle. In the stomach and small intestine it throws the lining into small folds, which adds surface area and keeps every cell exposed to the contents.

The other three layers

Layer	Main tissue	What it holds
Submucosa	Areolar connective tissue (OpenStax calls it dense); binds the mucosa to the muscularis	Larger blood and lymphatic vessels that receive absorbed food molecules; the submucosal plexus; glands in some regions
Muscularis	Smooth muscle in most of the tube, in two sheets: an inner circular layer and an outer longitudinal layer, with the myenteric plexus between them	Skeletal muscle instead in the mouth, pharynx, and the upper and middle esophagus (voluntary swallowing) and in the external anal sphincter (voluntary defecation)
Serosa	A serous membrane: simple squamous epithelium (mesothelium) over areolar connective tissue. It is the visceral peritoneum and suspends the tract in the abdominal cavity	Where the tube is not in the peritoneal cavity, an adventitia of areolar connective tissue replaces it: most of the esophagus, and the back of retroperitoneal organs

- Read the wall from the inside out: mucosa, submucosa, muscularis, serosa.
- The muscularis mucosae is a thin layer inside the mucosa. It is not the muscularis. Keep them in separate rows of your table.
- Two places break the muscle pattern. The stomach has a third, oblique layer (competency 115). The large intestine gathers its outer longitudinal layer into three bands, the teniae coli (competency 117).

Competency 110. Enteric nerve plexuses

In your course notes: Module 4 packet, Chapter 3, competency 110 (page 26).

Dr. Rennie's video for this part: Digestive system concept video, chapter Muscle layers and gut nerves, and Autonomic control and plexuses.

You should be able to: Locate the submucosal and myenteric plexuses in the gut wall and match each to the wall layer it sits within.

The tract has its own nervous system built into its wall, the enteric nervous system, sometimes called the brain of the gut. About 100 million neurons run from the esophagus to the anus, arranged in two plexuses. Each sits in a different layer, and the layer tells you the job.

Plexus	Wall layer	Motor neurons supply	Controls
Myenteric plexus (Auerbach)	Muscularis, between the inner circular and outer longitudinal layers	The two smooth muscle layers	Motility: how often and how hard the muscularis contracts
Submucosal plexus (Meissner)	Submucosa, just deep to the mucosa	The secretory cells of the mucosa and the vessels to it	Secretion by the GI organs, and blood flow to the lining

- Each plexus holds motor neurons, interneurons, and sensory neurons. Interneurons connect the two plexuses. Sensory neurons reach the mucosa, where chemoreceptors respond to chemicals in the food and mechanoreceptors respond to stretch when the wall fills.
- The name tells you the layer. Sub-mucosal is in the submucosa. Myenteric comes from myo, muscle, so it sits in the muscularis.
- Secretion goes with the submucosal plexus; movement goes with the myenteric plexus.

The autonomic nervous system adjusts the enteric system

- Parasympathetic fibers reach the tract through the vagus nerve (cranial nerve X) for everything down to about the first half of the large intestine, and through pelvic splanchnic nerves from the sacral cord for the last half. Preganglionic fibers synapse on neurons in the myenteric and submucosal plexuses; some reach the smooth muscle and glands directly. Parasympathetic activity increases secretion and motility.
- Sympathetic fibers come from the thoracic and upper lumbar cord. Postganglionic fibers synapse on plexus neurons and inhibit them. Sympathetic activity decreases secretion and motility, which is why anger, fear, and anxiety slow digestion.
- GI reflexes: receptors in the wall sense stretch or chemicals, signal neurons in the enteric system, the CNS, or the ANS, and those neurons turn GI glands and smooth muscle up or down.

Competency 111. Peritoneum and folds

In your course notes: Module 4 packet, Chapter 3, competency 111 (page 27).

Dr. Rennie's video for this part: Digestive system concept video, chapter Peritoneum and abdominal folds.

You should be able to: Identify the parietal and visceral peritoneum, the peritoneal cavity, and the five folds greater omentum, lesser omentum, falciform ligament, mesentery, and mesocolon and state what each fold connects.

The peritoneum is the largest serous membrane in the body: a layer of simple squamous epithelium (mesothelium) on areolar connective tissue. Like the pleura, it has two layers with a film of serous fluid between them so the organs slide without friction.

Parietal peritoneum
: Lines the walls of the abdominopelvic cavity.

Visceral peritoneum
: Covers the organs. On a canal organ it is the serosa.

Peritoneal cavity
: The thin space between the two layers, holding a little lubricating peritoneal fluid.

Ascites
: In liver disease, pancreatic disease, and some cancers, the peritoneal cavity fills with liters of fluid that often has to be drained.

Unlike the pleura, the peritoneum has large folds that weave between the organs. A fold is a double layer of peritoneum. The folds hold organs to each other and to the walls, store fat between their layers, and carry the blood vessels, lymphatic vessels, and nerves to the organs. Five are named in this course. For each one, learn its two ends.

Fold	What it connects	Other features
Greater omentum	Hangs from the greater curvature of the stomach and the first part of the duodenum, drapes down in front of the small intestine, then folds back up and attaches to the transverse colon	The largest fold. Folds on itself into four layers. A fatty apron that grows with weight gain and holds many lymph nodes that fight infections of the tract
Falciform ligament	Attaches the liver to the anterior abdominal wall and the diaphragm	The liver is the only organ attached to the anterior wall. Its free lower edge holds the ligamentum teres
Lesser omentum	From the lesser curvature of the stomach and the first part of the duodenum up to the liver	Its free edge carries the hepatic portal vein, the hepatic artery, and the common bile duct, with some lymph nodes
Mesentery	A fan from the posterior abdominal wall to the jejunum and ileum	Double-layered, full of fat, with the vessels, lymphatics, and lymph nodes of the small intestine between its layers
Mesocolon	Two separate folds from the posterior wall: the transverse mesocolon to the transverse colon and the sigmoid mesocolon to the sigmoid colon	Carries the vessels to those parts of the colon. With the mesentery, it holds the intestines loosely so they can move as they mix and push food

- Greater goes with the greater curvature and hangs low. Lesser goes with the lesser curvature and runs up to the liver. Meso means middle: the mesentery and mesocolon hold the intestines out from the back wall like a hammock.
- The falciform ligament reaches the front wall; the mesentery and mesocolon reach the back wall.
- On the cadaver, the greater omentum is the first thing you see when the abdomen is opened. Lift it toward the head to find the transverse colon on its back surface and the small intestine below.

What this means for patients

- Peritonitis is acute inflammation of the peritoneum. Microbes get in through a surgical or accidental wound, a ruptured appendix, or a perforated intestine, and the infection can be life-threatening.
- Inflamed peritoneal surfaces rubbing together is also peritonitis, painful but not usually dangerous. People on peritoneal dialysis are at risk.

Competency 112. Retroperitoneal organs

In your course notes: Module 4 packet, Chapter 3, competency 112 (page 28).

Dr. Rennie's video for this part: Digestive system concept video, chapter Peritoneum and abdominal folds.

You should be able to: Distinguish retroperitoneal organs from intraperitoneal ones and name the kidneys, pancreas, duodenum, and parts of the colon as retroperitoneal.

Intraperitoneal organ

Wrapped almost completely in visceral peritoneum and hanging into the cavity on a fold, so it can move: the stomach, jejunum, ileum, transverse colon, sigmoid colon, liver, and spleen.

Retroperitoneal organ

Lies on the posterior abdominal wall in the retroperitoneum, the region behind the parietal peritoneum, with peritoneum on its front surface only. It is not in the peritoneal cavity, has no mesentery, and is held firmly in place. Its back surface is covered by an adventitia.

Organ or part	Position	How you can tell
Stomach	Intraperitoneal	Hangs between the lesser and greater omenta
Duodenum	Retroperitoneal (except its first part)	Fixed to the back wall in a C around the head of the pancreas
Jejunum and ileum	Intraperitoneal	Hang on the mesentery
Ascending colon	Retroperitoneal	Fixed to the right posterior wall
Transverse colon	Intraperitoneal	Hangs on the transverse mesocolon
Descending colon	Retroperitoneal	Fixed to the left posterior wall
Sigmoid colon	Intraperitoneal	Hangs on the sigmoid mesocolon
Pancreas	Retroperitoneal (except the tip of the tail)	Lies across the back wall behind the stomach
Kidneys	Retroperitoneal	Against the back wall on either side of the spine

- The colon alternates: ascending fixed, transverse hanging, descending fixed, sigmoid hanging. The two parts with a mesocolon are the two intraperitoneal parts.
- SADPUCKER for the long list: Suprarenal (adrenal) glands, Aorta and inferior vena cava, Duodenum (except the first part), Pancreas (except the tail), Ureters, Colon (ascending and descending), Kidneys, Esophagus (the abdominal part), Rectum.

Competency 113. Mouth, pharynx, esophagus

In your course notes: Module 4 packet, Chapter 3, competency 113 (page 29).

Dr. Rennie's video for this part: Digestive system concept video, chapter Mouth and pharynx anatomy.

You should be able to: Identify the oral cavity boundaries, hard and soft palate, uvula, fauces, tonsils, the three pharyngeal regions, and the esophagus with its hiatus.

The mouth (oral cavity, buccal cavity)

Boundary	What forms it
Anterior	The lips (labia): fleshy folds of skin outside and nonkeratinized stratified squamous mucosa inside, with the orbicularis oris between. The labial frenulum, a midline fold, ties each lip to its gum
Lateral	The cheeks: skin outside, mucosa inside, the buccinator muscle between. Lips and cheeks keep food between the teeth and help with speech
Superior (roof)	The hard palate in front and the soft palate behind
Inferior (floor)	The tongue and the muscles under it (mylohyoid)
Posterior	The fauces, the opening into the oropharynx

Oral vestibule
: The space between the lips and cheeks on the outside and the gums and teeth on the inside.

Oral cavity proper
: The space from the gums and teeth back to the fauces.

Hard palate
: The bony front of the roof, formed by the palatine processes of the maxillae and the horizontal plates of the palatine bones under mucosa. It walls the mouth off from the nasal cavity, so you can chew and breathe at the same time, and gives the tongue a firm surface to press food against.

Soft palate
The muscular, arch-shaped back of the roof, between the oropharynx and nasopharynx. Its back border opens into the oropharynx through the fauces.

Uvula
The finger-shaped muscular tag hanging from the back edge of the soft palate. During a swallow the soft palate and uvula lift and seal the nasopharynx, so food and liquid do not enter the nose.

Palatoglossal arch
The front arch, running from the soft palate down to the base of the tongue.

Palatopharyngeal arch
The back arch, running from the soft palate to the side of the pharynx.

Palatine tonsils
Between the two arches on each side. These are the tonsils you see in an open mouth.

Lingual tonsil
On the back (root) of the tongue.

The pharynx

A funnel-shaped tube of skeletal muscle lined with mucous membrane, from the choanae to the esophagus behind and the larynx in front. Three regions: the nasopharynx (air only), the oropharynx, and the laryngopharynx (air and food). The pharyngeal tonsil sits in the nasopharynx. A swallowed bolus goes mouth, oropharynx, laryngopharynx, esophagus, stomach, pushed by the pharyngeal constrictors. Breathing stops for the moment the bolus passes.

The esophagus

- A collapsible muscular tube about 25 cm (10 inches) long. It starts at the lower end of the laryngopharynx at the level of the cricoid cartilage, runs through the mediastinum behind the trachea and in front of the vertebral column, pierces the diaphragm, and joins the stomach at the cardia.
- The esophageal hiatus is the opening in the diaphragm it passes through, at about T10.
- It secretes mucus and carries food to the stomach. No digestion and no absorption happen here.

Layer	In the esophagus
Mucosa	Nonkeratinized stratified squamous epithelium (wear and tear), lamina propria, muscularis mucosae. Mucous glands near the stomach
Submucosa	Areolar connective tissue, vessels, mucous glands
Muscularis	Upper third skeletal muscle, middle third mixed, lower third smooth muscle. Two sphincters: the upper esophageal sphincter (skeletal) and the lower esophageal (cardiac) sphincter (smooth), near the heart
Adventitia	Areolar connective tissue with no mesothelium, blending into the mediastinum to anchor the esophagus. Only the short abdominal segment has a serosa

What this means for patients

- Hiatal hernia: part of the stomach pushes up through the esophageal hiatus into the thorax. It weakens the support the diaphragm gives the lower esophageal sphincter and often comes with reflux (competency 119).

Competency 114. Major canal sphincters

In your course notes: Module 4 packet, Chapter 3, competency 114 (page 31).

You should be able to: Locate the upper and lower esophageal, pyloric, ileocecal, and internal and external anal sphincters and state which regions each one separates.

A sphincter is a ring of muscle around the tube that can squeeze it shut. Sphincters sit at borders between regions. They control when contents move on and most keep contents from flowing backward. Learn each one by the two regions on either side of it.

Sphincter	Where	Separates	Muscle
Upper esophageal sphincter	Top of the esophagus, at the cricoid cartilage	Laryngopharynx from esophagus	Skeletal; stays closed between swallows so air does not enter the esophagus
Lower esophageal sphincter (also called the cardiac sphincter or gastroesophageal sphincter)	Bottom of the esophagus, just under the diaphragm	Esophagus from stomach (cardia)	Smooth; the diaphragm around the hiatus helps hold it shut, keeping acid out of the esophagus
Pyloric sphincter	Around the pylorus at the stomach exit	Stomach from duodenum	Smooth, a thickening of the circular layer; meters small amounts of chyme into the duodenum
Ileocecal sphincter	End of the ileum where it opens into the cecum	Small intestine (ileum) from large intestine (cecum)	Smooth; with folds of mucosa it forms the ileocecal valve, which keeps colon contents out of the ileum
Internal anal sphincter	Around the anal canal, inside the external sphincter	Anal canal from the outside	Smooth, involuntary
External anal sphincter	Around the anal canal, outside the internal sphincter	Anal canal from the outside	Skeletal, voluntary

- Only the two ends of the canal have skeletal sphincters: the upper esophageal at the top and the external anal at the bottom. Everything in between is smooth muscle.
- The sphincter of Oddi (hepatopancreatic sphincter) is on your lab list with the small intestine, but it is not a sphincter of the canal. It controls bile and pancreatic juice entering the duodenum at the major duodenal papilla (competencies 123 and 124).

Competency 121. Salivary glands

In your course notes: Module 4 packet, Chapter 4, competency 121 (page 40).

Dr. Rennie's video for this part: Digestive system concept video, chapter Mouth and pharynx anatomy.

You should be able to: Locate the parotid, submandibular, and sublingual glands and state where the duct of each opens.

Saliva keeps the mouth and pharynx moist, cleans the mouth and teeth, and during a meal lubricates and dissolves food and starts the chemical breakdown of starch. Small glands in the lining (labial, buccal, palatal, and lingual glands) keep the mouth wet all the time. Most saliva comes from three pairs of major glands outside the lining, each sending its saliva in through a duct.

Gland	Location	Where the duct opens	Saliva
Parotid gland	In front of and below the ear, between the skin and the masseter muscle; the largest	The parotid duct crosses the masseter, pierces the buccinator, and opens into the vestibule opposite the upper second molar	Watery (serous), with salivary amylase
Submandibular gland	Floor of the mouth, along the inner surface of the mandible	The submandibular duct runs forward under the mucosa of the floor and opens just lateral to the lingual frenulum	Mostly serous with some mucus, a little thicker
Sublingual gland	Under the tongue in the floor of the mouth, above the submandibular gland; the smallest	Many small sublingual ducts opening along the floor of the mouth	Mostly mucus, thick, very little amylase

- Each name tells you the spot. Parotid, beside the ear. Submandibular, below the mandible. Sublingual, below the tongue.
- Only the parotid duct opens in the cheek. The other two open under the tongue.
- Saliva is about 99.5 percent water. The rest is ions (sodium, potassium, chloride, bicarbonate, phosphate), dissolved gases, and organic substances: urea, uric acid, mucus, IgA, lysozyme, and salivary amylase, which splits starch into maltose, maltotriose, and alpha-dextrins.

What this means for patients

- Mumps is a paramyxovirus infection that targets the parotid glands. The glands swell on one or both sides in front of the ramus of the mandible, with fever, malaise, and throat pain on swallowing or with sour foods. In about 30 percent of males past puberty the testes become inflamed; sterility is rare unless both sides are involved. A vaccine prevents it.

Competency 120. Teeth and tongue

In your course notes: Module 4 packet, Chapter 4, competency 120 (page 38).

Dr. Rennie's video for this part: Digestive system concept video, chapter Mouth and pharynx anatomy.

You should be able to: Identify the teeth and tongue with the lingual frenulum, papillae, intrinsic and extrinsic muscles, and lingual glands.

The tongue

The tongue is skeletal muscle covered with mucous membrane. With its muscles it forms the floor of the oral cavity. A median septum divides it into right and left halves along its whole length, and it attaches below to the hyoid bone, the styloid process of the temporal bone, and the mandible.

Muscle group	Origin and insertion	Action
Extrinsic (genioglossus, hyoglossus, styloglossus, palatoglossus)	Start on bone or the soft palate outside the tongue, end in the connective tissue of the tongue	Move the whole tongue: side to side, in and out, up and down, to position food for chewing, shape the bolus, and push it back to swallow. They also hold the tongue in place
Intrinsic (superior and inferior longitudinal, transverse, vertical)	Start and end inside the tongue	Change the shape and size of the tongue for speech and swallowing: curl, flatten, narrow, shorten

- The hypoglossal nerve (cranial nerve XII) supplies all the tongue muscles except palatoglossus, which the vagus supplies.
- Extrinsic starts outside the tongue and moves it. Intrinsic sits within it and changes its shape.

Lingual frenulum

The midline fold of mucosa under the tongue that ties it to the floor of the mouth and limits how far back it can move. A short or rigid frenulum is ankyloglossia (tongue-tie), which impairs speech and can be corrected surgically.

Lingual papillae

Projections of the lamina propria covered with stratified squamous epithelium on the top and sides of the tongue. Some carry taste buds; all give the tongue grip on food.

Lingual glands

Small glands in the lamina propria of the tongue. They release mucus and a watery fluid with lingual lipase, which starts the breakdown of dietary triglycerides into fatty acids and diglycerides.

Papillae	Where and what they look like	Taste buds
Filiform	Thin, pointed, the most numerous, over most of the top surface	No. They give the tongue friction
Fungiform	Mushroom-shaped, scattered among the filiform, most on the tip and sides	Yes
Vallate (circumvallate)	Large, round, about 8 to 12 in a V-shaped row near the back, each in a trench	Yes
Foliate	Leaf-like folds on the sides toward the back	Yes

The teeth

- Each tooth sits in a socket (an alveolus) in the maxilla or mandible, held by a periodontal ligament. The joint is a gomphosis, a fibrous joint.
- Children have 20 deciduous teeth. Adults have 32 permanent teeth: 8 incisors (cut), 4 canines (tear), 8 premolars (crush), and 12 molars (grind), including the four third molars (wisdom teeth).

Crown, neck, root

The crown is above the gum (gingiva), the neck is at the gum line, and the root sits in the socket.

Enamel

Covers the crown. The hardest substance in the body.

Dentin

Forms most of the tooth, under the enamel.

Pulp cavity and root canal

The center of the tooth, holding the pulp (vessels, nerves, connective tissue). It continues down each root as the root canal.

Cementum

Covers the root and anchors it to the periodontal ligament.

Worked questions

Nine questions in the style of the lecture exam. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 109

Which layer of the gut wall contains the large blood vessels, lymphatic vessels, and the submucosal plexus?

A. Mucosa

B. Submucosa

C. Muscularis

D. Serosa

Answer: B. The submucosa is the connective tissue layer between the mucosa and the muscularis. It carries the larger vessels that receive absorbed nutrients and holds the submucosal plexus.

Why the others are wrong

- **A** The mucosa has small capillaries and lymphatic tissue in its lamina propria, but the large vessels and the plexus are one layer out.
- **C** The muscularis is the two smooth muscle layers. The plexus inside it is the myenteric plexus, not the submucosal.
- **D** The serosa is the thin outer serous membrane, the visceral peritoneum. It is not where the vessels run.

QUESTION 2 · COMPETENCY 109

Which region of the canal is lined with simple columnar epithelium?

A. Esophagus

B. Oropharynx

C. Stomach

D. Anal canal

Answer: C. The stomach and both intestines secrete and absorb, so they are lined with simple columnar epithelium.

Why the others are wrong

- **A** The esophagus carries rough food past, so its lining is stratified squamous for protection.
- **B** The oropharynx carries food and air and is stratified squamous.
- **D** The anal canal takes friction at the exit of the tube and is stratified squamous near the anus.

QUESTION 3 · COMPETENCY 110

The myenteric plexus is found

A. in the lamina propria of the mucosa

B. in the submucosa, just under the mucosa

C. between the inner circular and outer longitudinal layers of the muscularis

D. between the muscularis and the serosa

Answer: C. Myenteric means in the muscle. It sits between the two smooth muscle layers it controls, and it runs motility.

Why the others are wrong

- **A** The lamina propria holds vessels and MALT, not a plexus.
- **B** That is the submucosal plexus, which runs secretion and blood flow to the lining.
- **D** Nothing named in this course sits between the muscularis and the serosa. Both plexuses are inside the wall.

QUESTION 4 · COMPETENCY 111

Which peritoneal fold attaches the liver to the anterior abdominal wall and the diaphragm?

A. Lesser omentum

B. Greater omentum

C. Falciform ligament

D. Mesentery

Answer: C. The falciform ligament is the sickle-shaped fold from the front of the liver to the anterior wall and diaphragm. The liver is the only organ attached to the anterior wall.

Why the others are wrong

- **A** The lesser omentum connects the stomach and the first part of the duodenum to the liver. It does not reach the body wall.
- **B** The greater omentum hangs from the greater curvature of the stomach and attaches to the transverse colon.
- **D** The mesentery holds the jejunum and ileum to the posterior wall.

QUESTION 5 · COMPETENCY 112

Which of these is retroperitoneal?

A. Transverse colon

B. Jejunum

C. Stomach

D. Descending colon

Answer: D. The descending colon is fixed to the left posterior wall behind the parietal peritoneum. Ascending and descending colon are the two retroperitoneal parts.

Why the others are wrong

- **A** The transverse colon hangs on the transverse mesocolon, so it is intraperitoneal.
- **B** The jejunum hangs on the mesentery, wrapped in visceral peritoneum.
- **C** The stomach hangs between the lesser and greater omenta inside the peritoneal cavity.

QUESTION 6 · COMPETENCY 113

During a swallow, which structure lifts to seal off the nasopharynx?

A. The epiglottis

B. The soft palate and uvula

C. The hard palate

D. The palatine tonsils

Answer: B. The soft palate is muscle, so it can move. It and the uvula rise and press against the back wall of the pharynx, closing the nasopharynx so food does not enter the nose.

Why the others are wrong

- **A** The epiglottis tips down over the laryngeal inlet to keep food out of the airway below, not the nose above.
- **C** The hard palate is bone and cannot move.
- **D** The palatine tonsils are lymphatic tissue between the arches. They guard the throat; they do not seal anything.

QUESTION 7 · COMPETENCY 114

Which sphincter is skeletal muscle under voluntary control?

A. Lower esophageal sphincter

B. Pyloric sphincter

C. Internal anal sphincter

D. External anal sphincter

Answer: D. The external anal sphincter is skeletal muscle, the one you control. Only the two ends of the canal, the upper esophageal and the external anal sphincters, are skeletal.

Why the others are wrong

- **A** The lower esophageal sphincter is smooth muscle helped by the diaphragm.
- **B** The pyloric sphincter is a thickening of the circular smooth muscle layer.
- **C** The internal anal sphincter is smooth muscle and involuntary. It sits inside the external one.

QUESTION 8 · COMPETENCY 121

The parotid duct opens

A. beside the lingual frenulum

B. along the floor of the mouth under the tongue

C. inside the cheek opposite the upper second molar

D. at the base of the tongue

Answer: C. The parotid duct runs forward across the masseter, pierces the buccinator, and opens into the vestibule opposite the upper second molar.

Why the others are wrong

- **A** That is the submandibular duct.
- **B** That is where the many small sublingual ducts open.
- **D** The base of the tongue holds the lingual tonsil, not a salivary duct opening.

QUESTION 9 · COMPETENCY 120

Which papillae have no taste buds?

A. Fungiform

B. Vallate

C. Foliate

D. Filiform

Answer: D. Filiform papillae are the thin, pointed, most numerous papillae. They give the tongue friction and carry no taste buds.

Why the others are wrong

- **A** Fungiform papillae on the tip and sides carry taste buds.
- **B** Vallate papillae, the big ones in the V at the back, carry taste buds.
- **C** Foliate papillae on the sides carry taste buds.